



Rail Transit Noise & Vibration

Derek Watry – Technical Director

Wilson Ihrig / RWDI

B O S T O N

Rail Transit Noise & Vibration



- Objective: Teach you a few things about rail noise and vibration
- Not objective: Make you rail noise and vibration experts



Fundamentals



Sound: Something is moving / vibrating



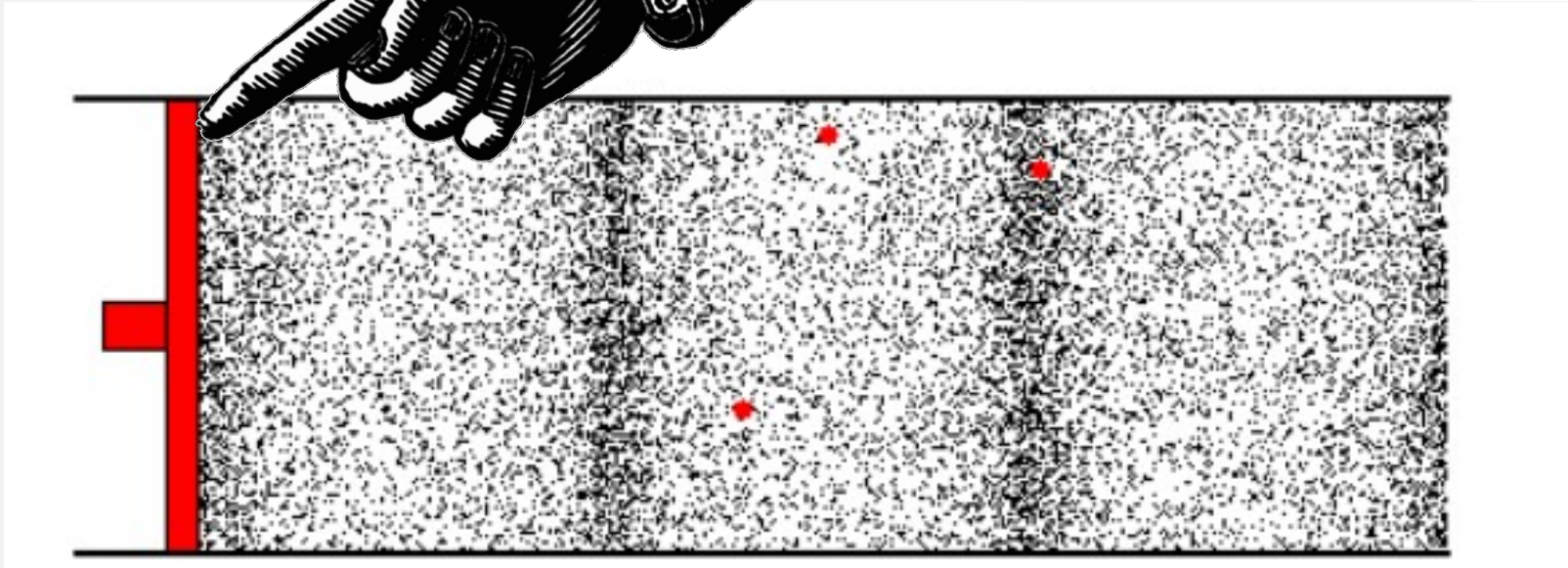
Fundamentals

• Something is moving / vibrating → Next comes wave propagation

- Air: Sound
- Ground: Vibration
- Water: Who knows?



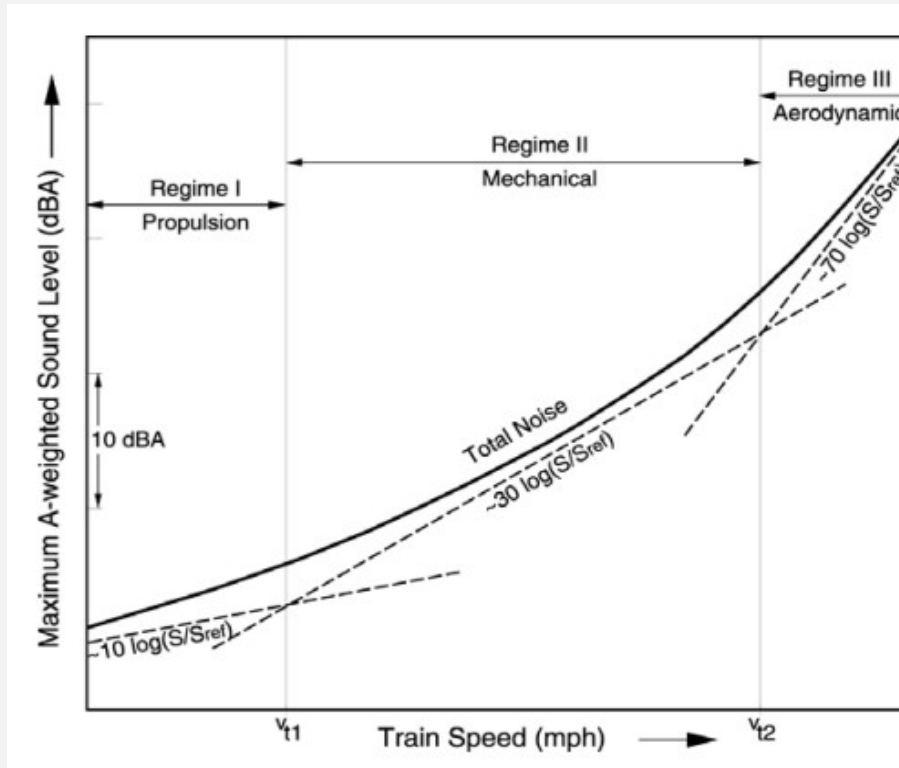
Fundamentals



Cool illustration of acoustic propagation from the University of Southampton (UK)

Sources of Rail Transit Noise

- Foreshadowing: Wheel/Rail Interaction
- HVAC & ancillary
- Traction / chopper
- Bells, horns
- Aerodynamic





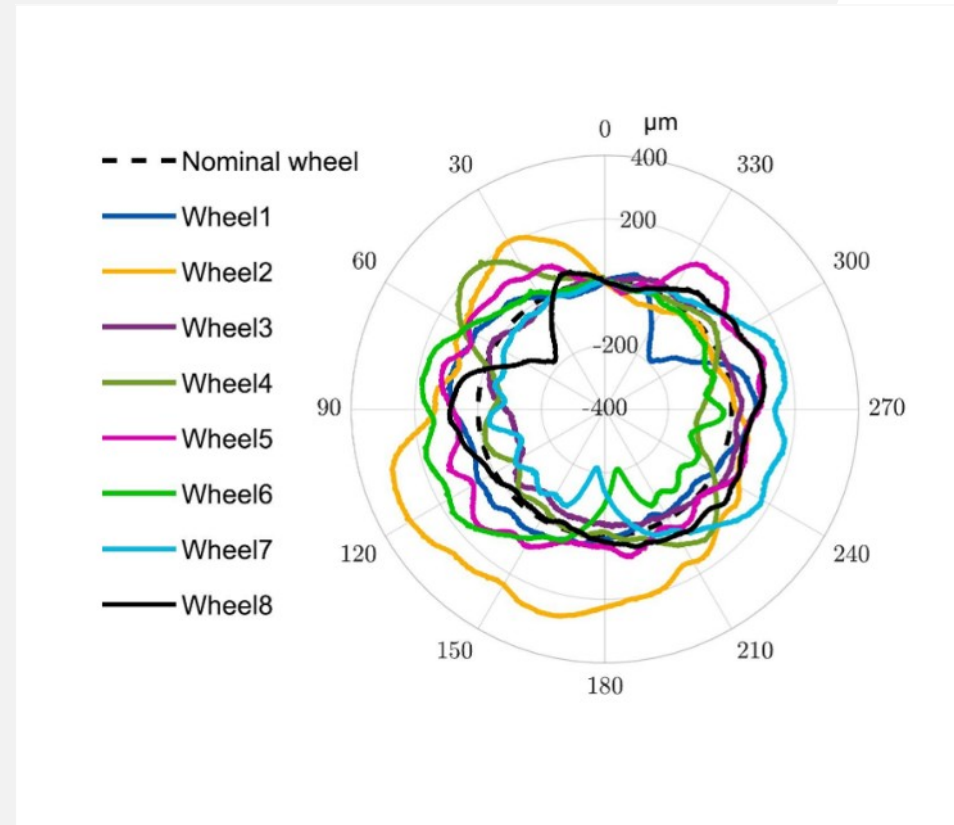
Wheel/Rail Noise Sources

- Corrugation
- Wheel flats and spalls
- Rail squats and spalls
- Special trackwork
- Jointed rail
- Wheel/rail roughness
- Curve squealing



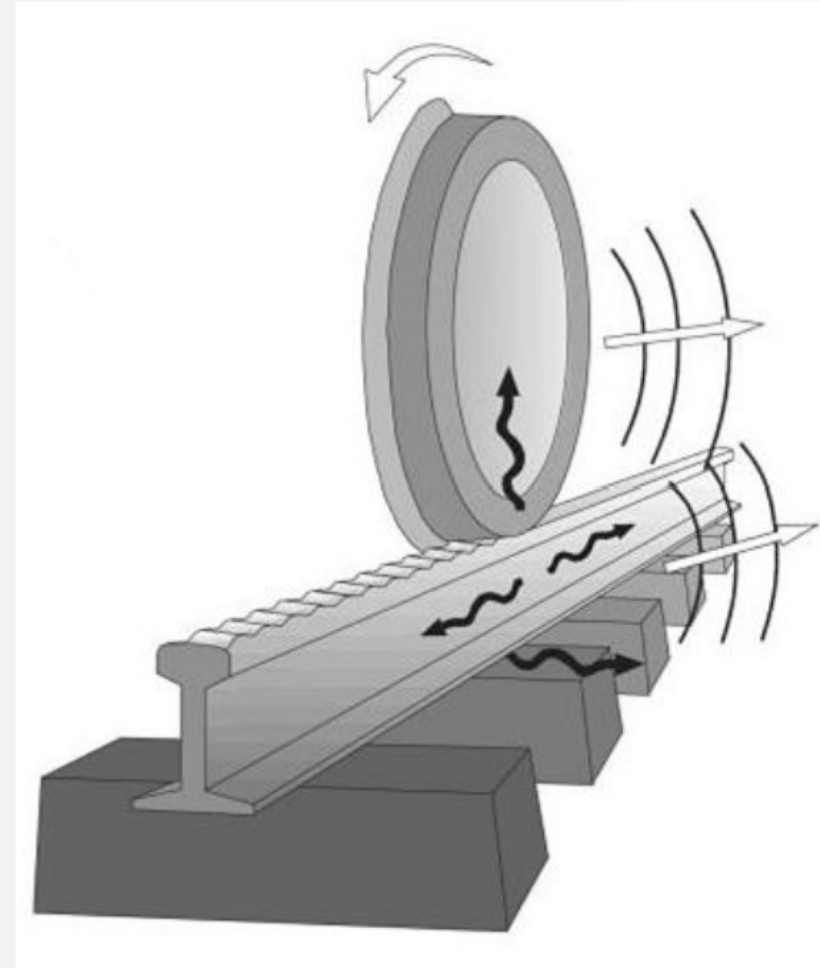
Wheel/Rail Vibration Sources

- Corrugation
- Wheel flats and spalls
- Rail squats and spalls
- Special trackwork
- Jointed rail
- Wheel/rail roughness
- Track fixation stiffness
- Vehicle design
- Wheel out of roundness



Propagation - Noise

- Vibrating wheels and rails radiate noise through the air: **Sound**



Propagation - Vibration

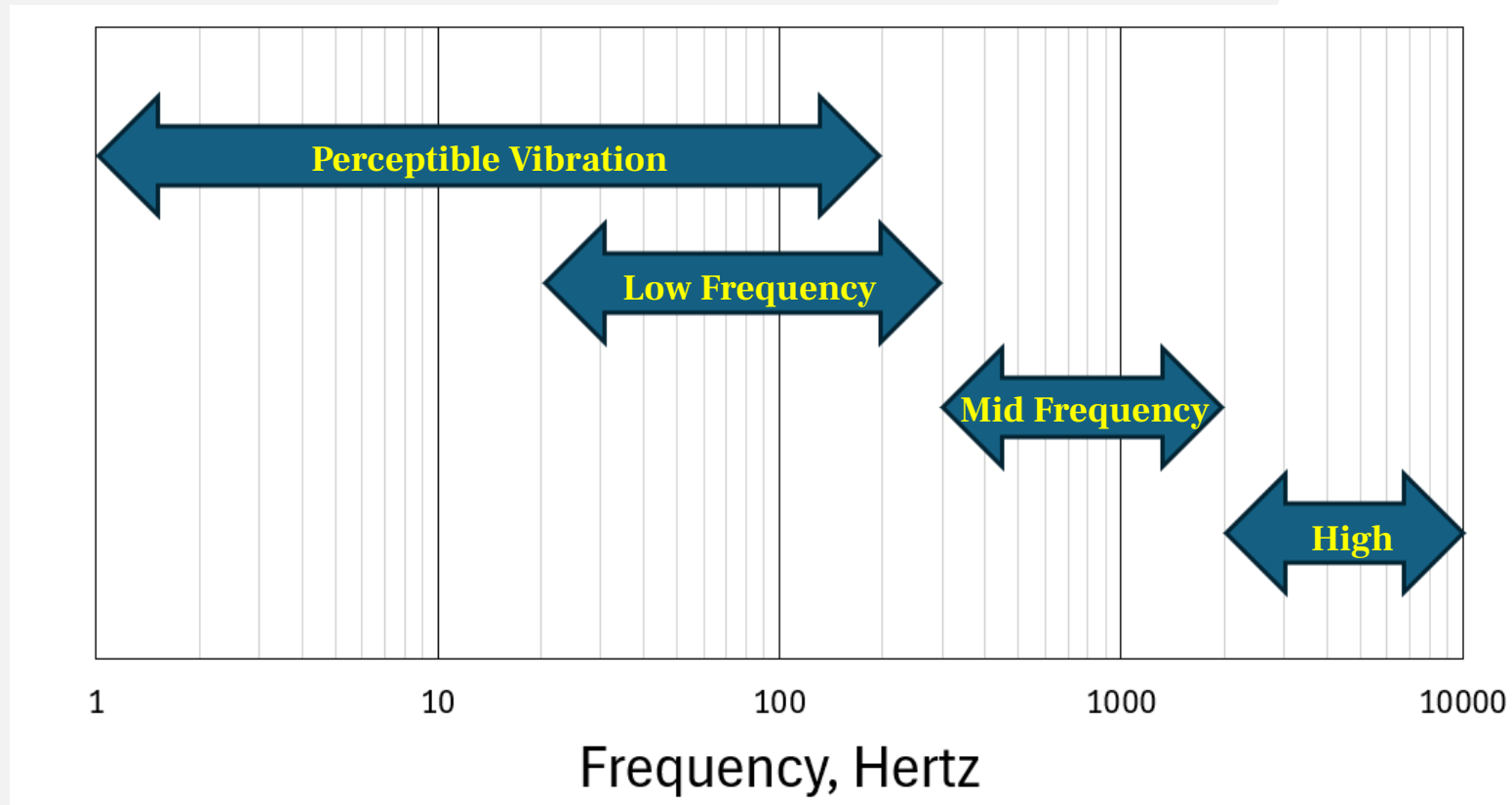
- Vibration propagation is more involved:
 - Track fixation
 - Tunnel structure
 - Ground (complicated)
 - Bldg foundation
 - Bldg structure
- Vibration
- Groundborne noise
(surfaces as speakers)





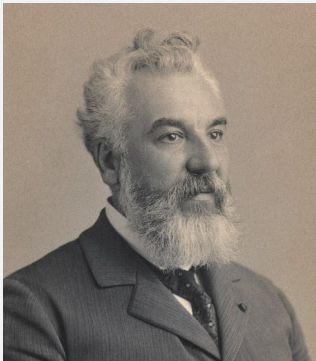
Measuring Noise & Vibration

- Two main “dimensions”: Frequency and Level
- Frequency



Measuring Noise & Vibration

- Level: Decibels – The World's Most Confusing Unit!



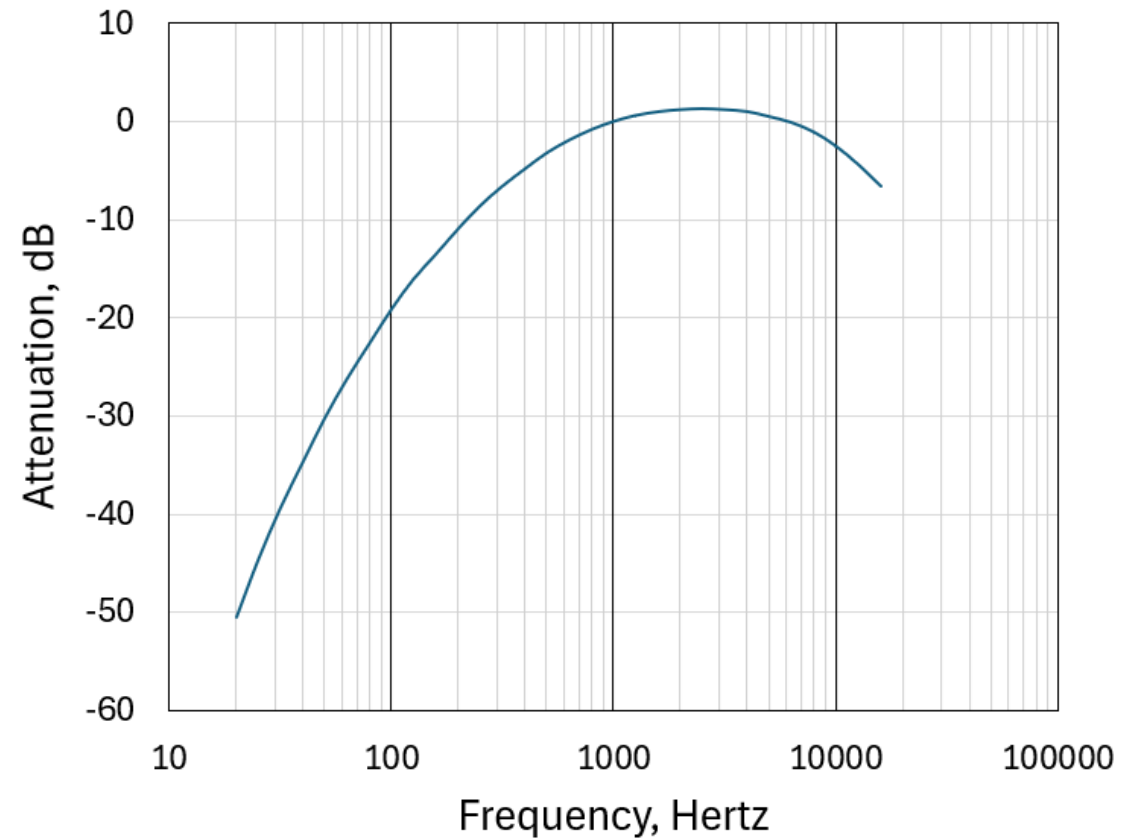
Alexander Graham Bell

$$\text{dB} = 10 \log\left(\frac{\text{power}}{\text{reference power}}\right)$$

$$60 \text{ dB} + 60 \text{ dB} = 63 \text{ dB !}$$

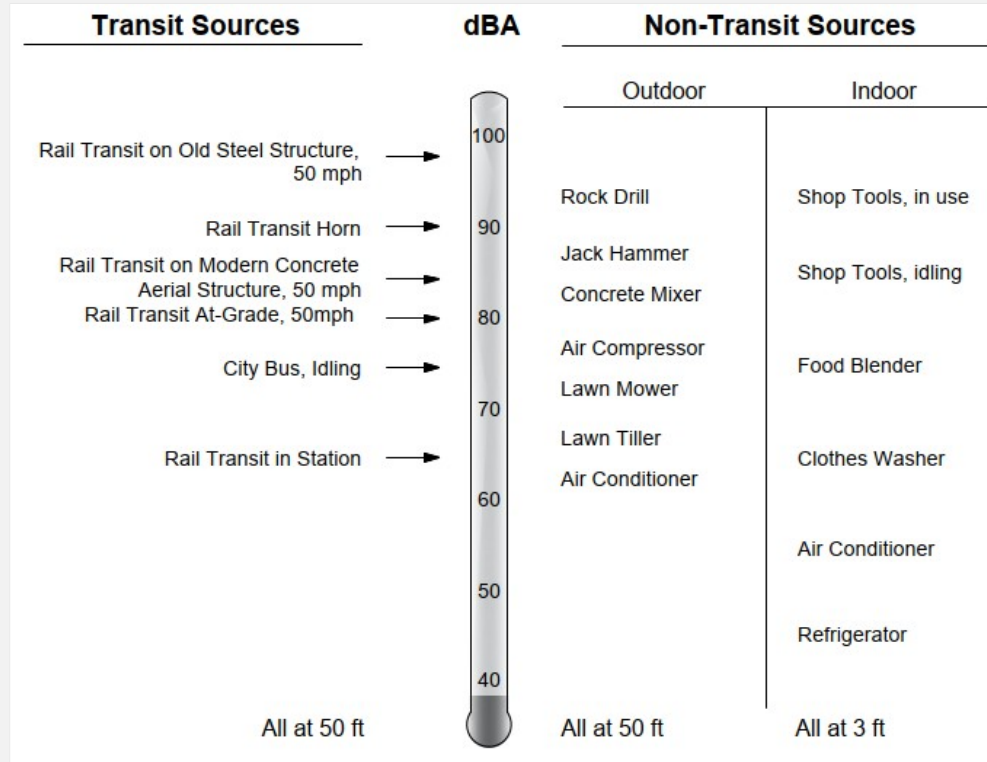
Measuring Noise

- Typical hearing response
 - A-weighting curve
 - Essentially a filter
- A-weighted decibel is a single number level that reflects how humans judge loudness
- Denoted by “dBA”



Measuring Noise

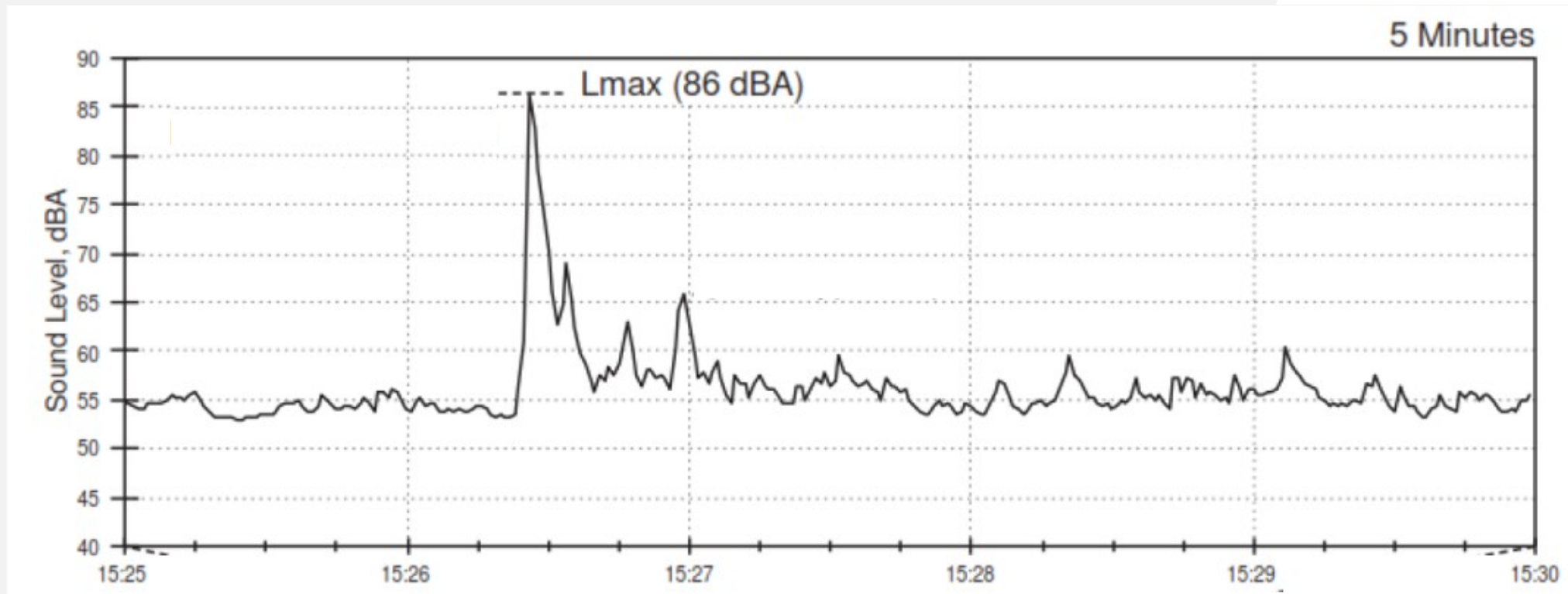
- A-weighted levels



dBA	Example	
130	Threshold of pain	Intolerable Extremely Noisy
120	Heavy rock concert	
110	Grinding on steel	Very noisy
100	Loud car horn at 3m	
90	Construction pneumatic hammering	Loud
80	Curbside of busy street	
70	Loud radio or television	Moderate to quiet
60	Department store	
50	General Office	Quiet to very quiet
40	Inside private office	
30	Inside bedroom	Almost silent
20	Recording studio	

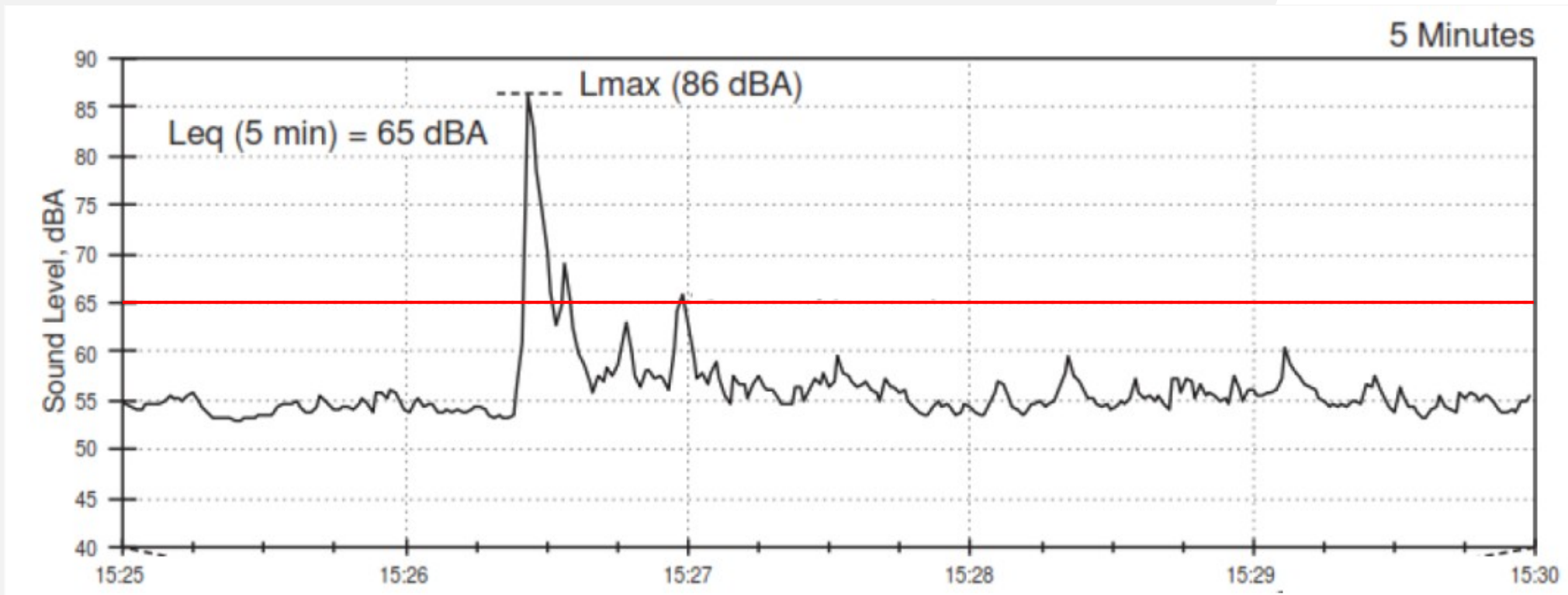
Measuring Noise & Vibration

- Accounting for time: Maximum Level, $L_{\max}$



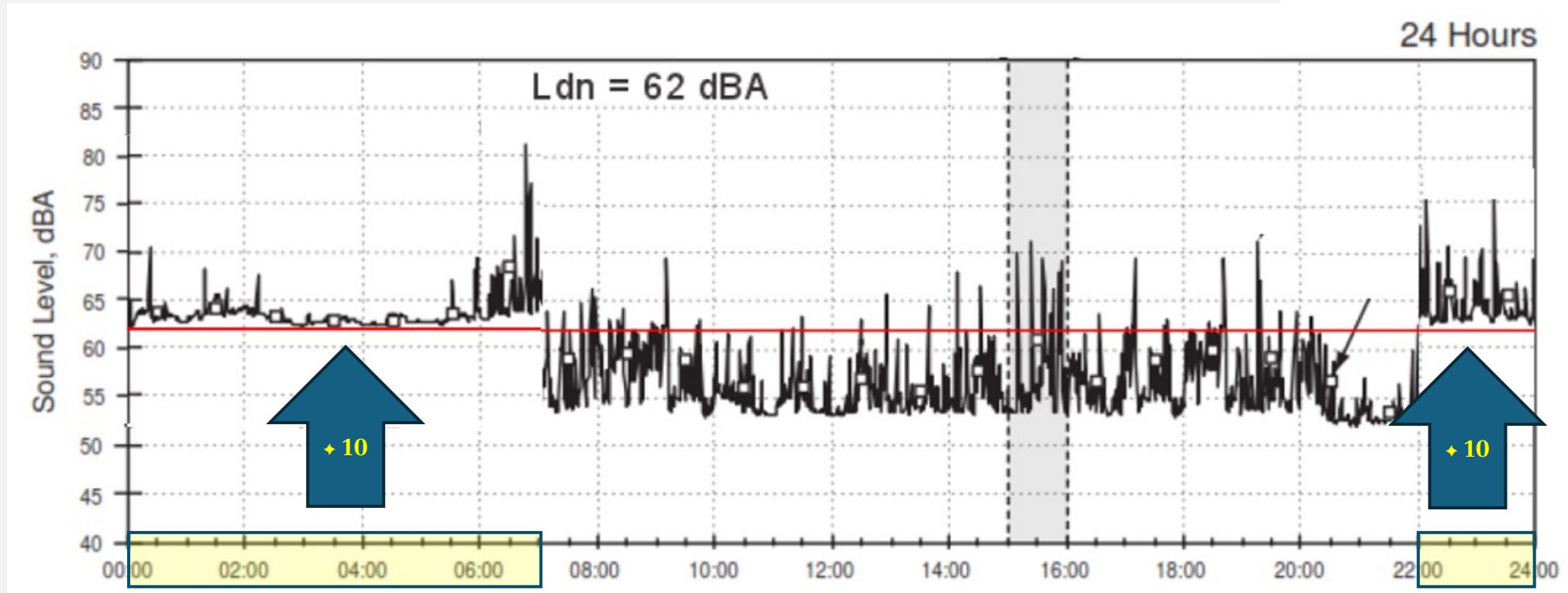
Measuring Noise & Vibration

- Accounting for time: Equivalent Level, $L_{eq}(\text{time})$



Measuring Noise

- Accounting for time: Day-Night Level, L_{dn}



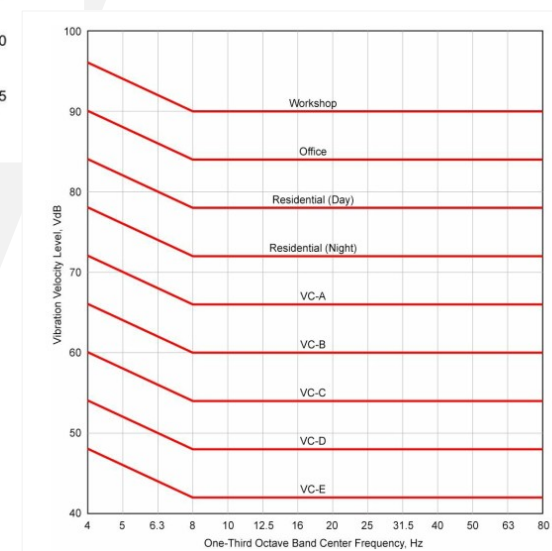
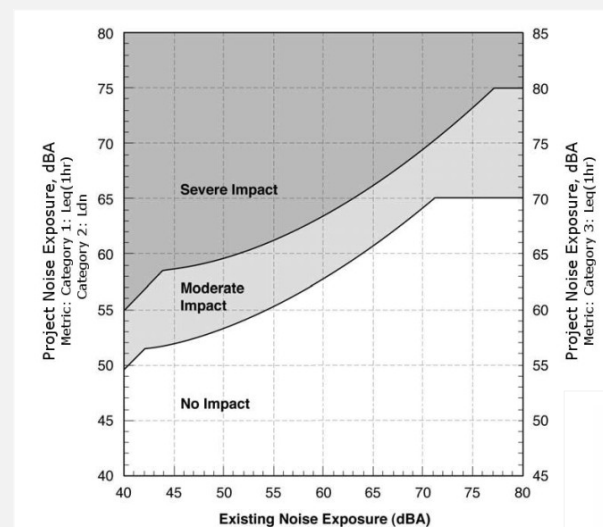
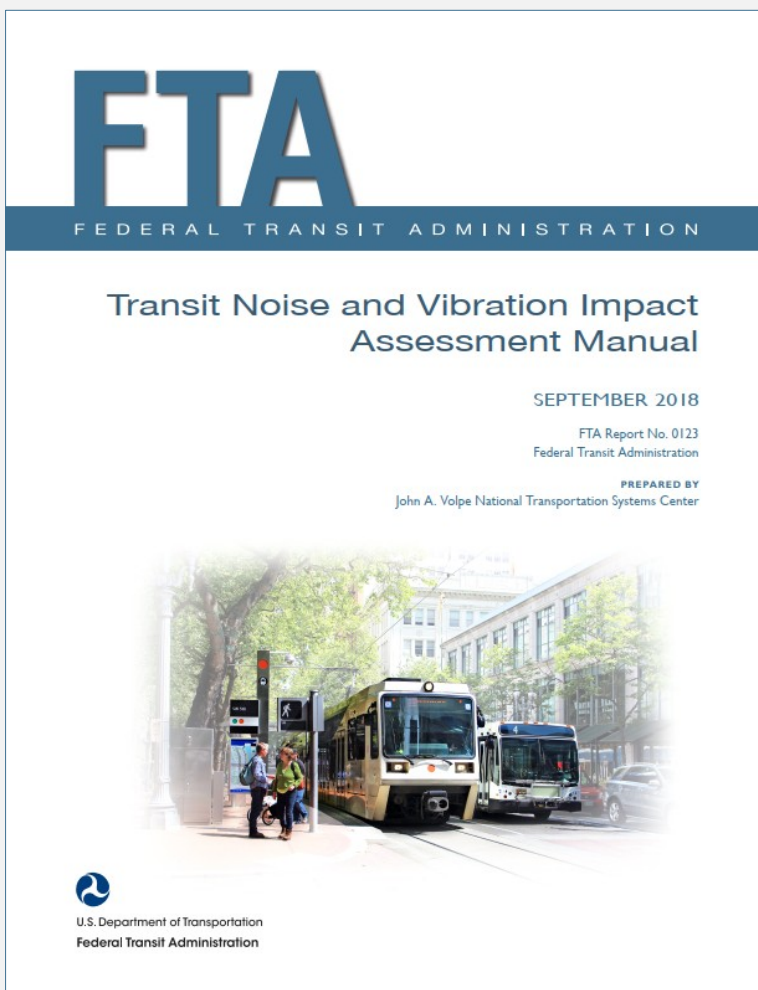
Measuring Noise & Vibration



- Accounting for time: Summary
 - Maximum Level, L_{max} - As bad as it gets
 - Equivalent Level, $L_{eq}(t)$ - The average over the given time period
 - Day-Night Level, L_{dn} - 24-hour average that weights nighttime noise
- Many other metrics



Assessing Noise & Vibration





What to do?

- Design – Better to prevent than correct
- Retrofit – What would life be with second chances?
- Maintenance – Necessary



Control Through Design – Track (1 of 3)

- Avoid tight-radius curves – $\text{Max}\{1000 \text{ ft}, 100 \times \text{wheelbase}\}$

- Gauge / super-elevation
Squealing up to 106 dBA

- Hard rails – Inhibit

- Consider track reso

- Incorporate flat det



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Control Through Design – Track (2 of 3)



- Special trackwork – Locate with due consideration
- Low-impact frogs – Min gaps, spring, movable point, jump
- Jointed rail – Avoid if possible
- Track absorption – May also be retrofit



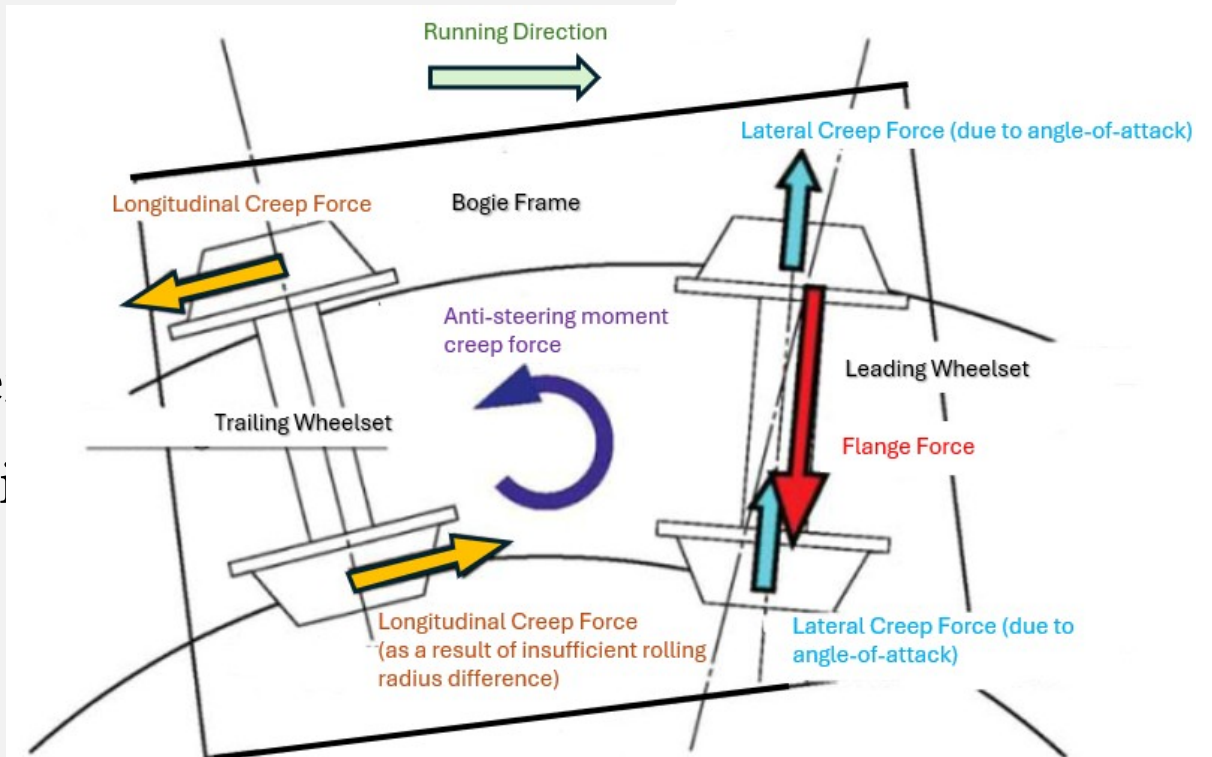
Control Through Design – Track (3 of 3)

- Track fixation stiffness
 - Soft rail pads
 - Soft fasteners
 - Under-tie pads
 - Booted tie
 - Ballast mat
 - Tire-derived aggregate
 - Floating slab



Control Through Design – Vehicle

- Curve/flanging wheel squeal
 - Resilient wheels
 - Control steering angle of attack
- Wheel flats, spalls, and roughness
 - Anti-lock brakes / wheel slide prevention
 - Operator training – Anticipatory driving
- Truck design
 - Minimize unsprung mass
 - Soft primary suspension



Control Through Design – Structural

- Heavier tunnel, invert, slab, aerial structure
- Grout soft soil around tunnels
- Design “outside the box”
- Or just build a box





Retrofit Solutions

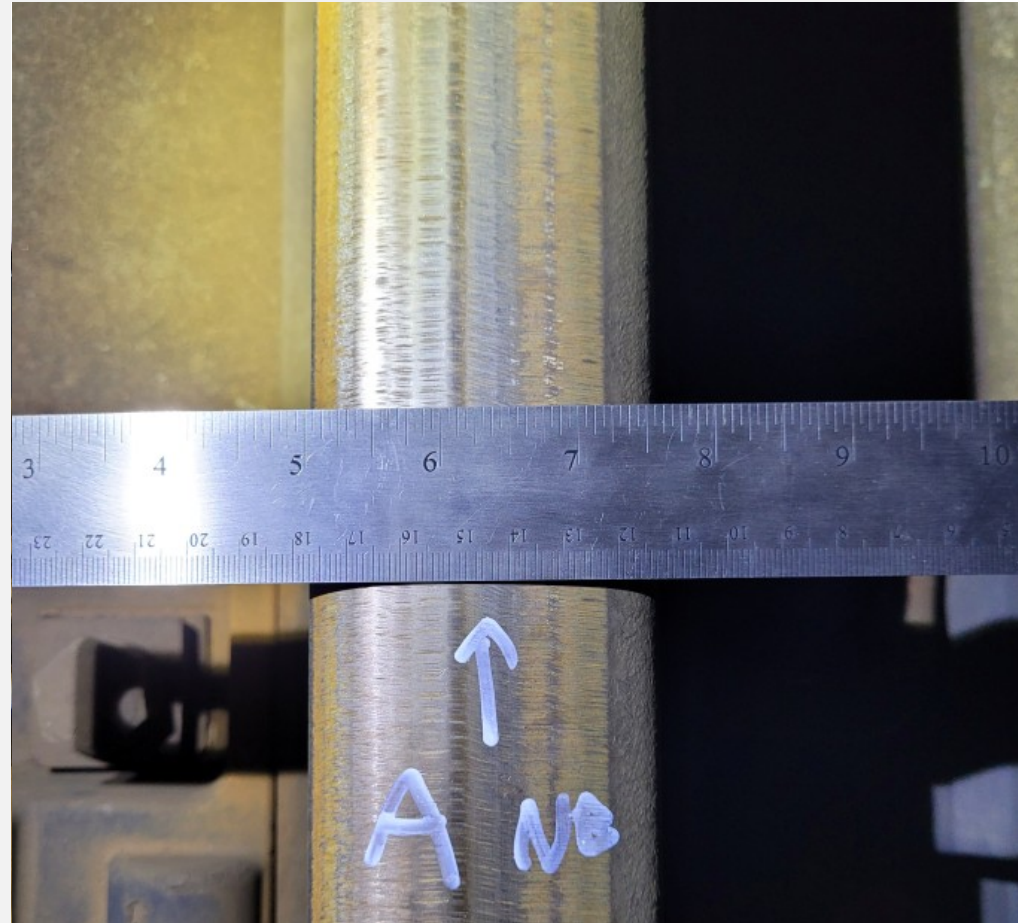


- Rolling noise – Rail dampers
- Corrugation
 - Optimize wheel/rail profiles
 - Friction modifiers
- Curve/flange squealing
 - Lubrication – Trackside, Vehicle
 - Wheel dampers
 - Top rail friction management
- Wheel flats and spalls
 - Friction management



Maintenance

- True wheels
 - Wheel flats and spalls
 - Out of roundness
- Grind rail
 - Corrugation
 - Rough rail





Questions & Answers



Thank you!